

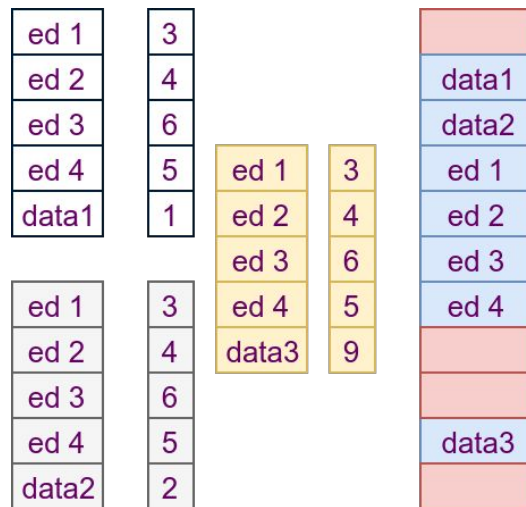
Paging Contd. and Segmentation

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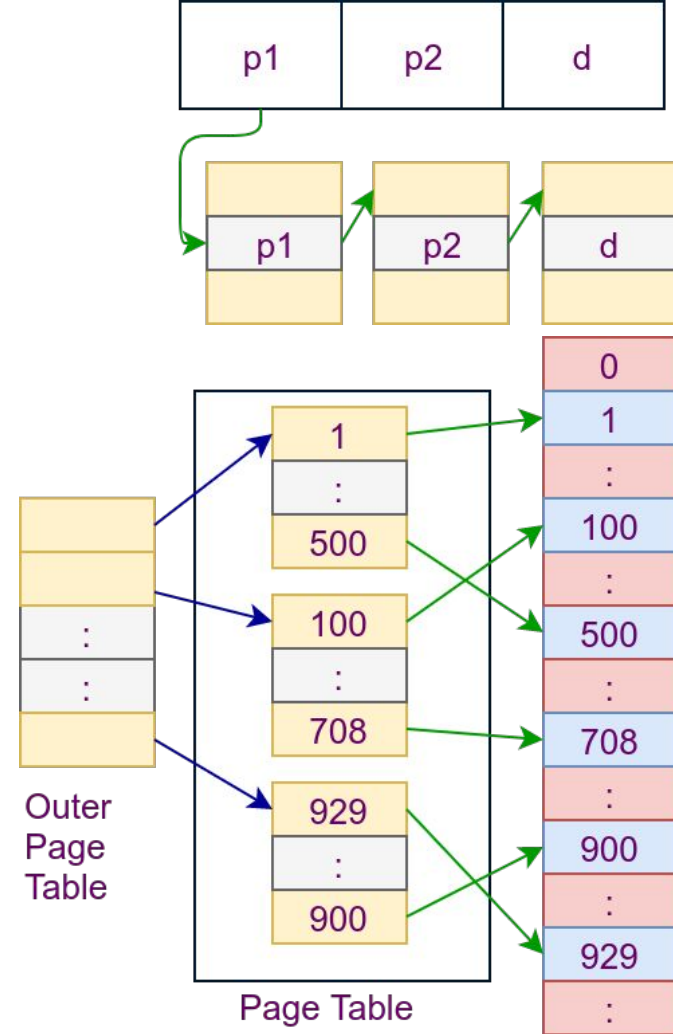
Shared Pages

- Advantage of paging -> sharing common code -> time sharing environment
 - Consider a system of 40 users -> each text editor
- Reentrant (multi-instance) code is a reusable routine that multiple programs can invoke, interrupt, and reinvoke simultaneously.
 - Non-self-modifying code - never changes during execution
 - Multiple process can use but data may be different
- One copy of such codee, text editor
 - Compilers, run-time libraries, DBMS
- Similar to sharing address spaces -
- Shared memory IPC - > Shared pages by few OS



Hierarchical Paging

- There are few common structure of Page Table
- Modern OS -> Large logical space (2^{32} to 2^{64})
 - If a page size 4KB (2^{12}) -> 1 million entries in Page Table
 - Each entry 4 bytes -> each process 4MB
- Divide the page table into small pieces
- Two way paging - Page table is paged
- Logical address - index to outer page, displacement within the page of outer page and page offset -> Forward-mapped page table
- Vax architecture - Section, Page, Offset
- In 64bit logical address system two-level paging becomes inappropriate
 - Another outer page of Outer page table



Inverted Page Table

Pages	Frames
0	OS
1	P1 p2
2	P2 p0
3	P1 p3
4	P2 p1
5	P1 p6
6	P1 p4
7	P2 p2

Inverted Page Table

Pages	Frames
0	X
1	X
2	F1
3	F3
4	F6
5	X
6	F5

Page Table of P1

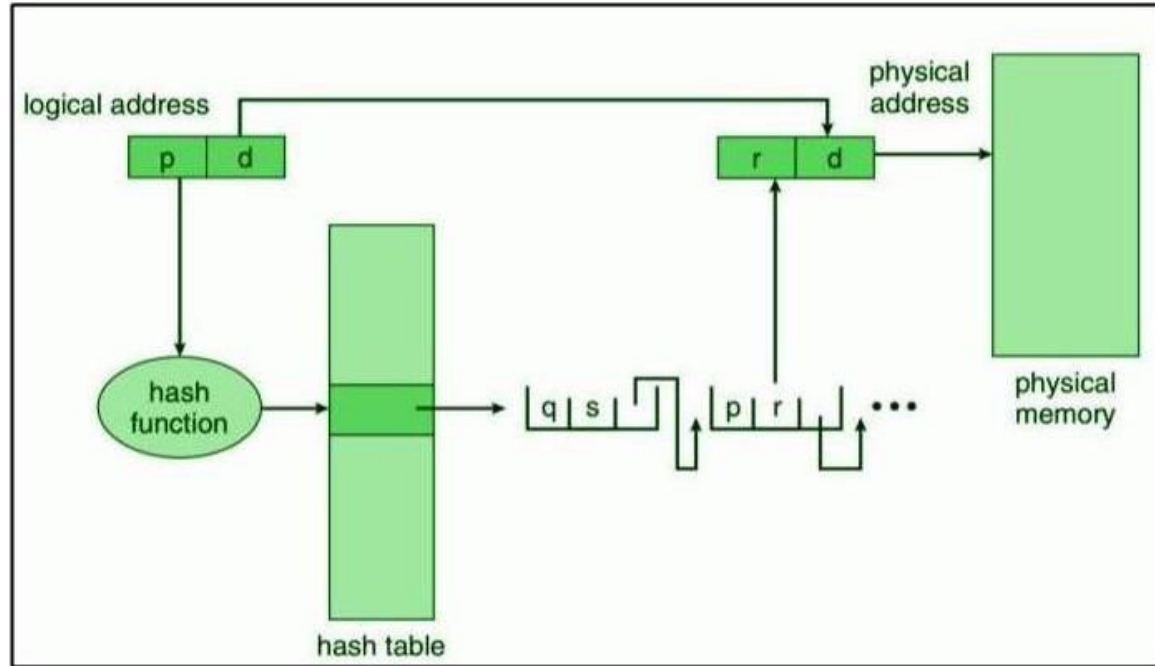
Pages	Frames
0	F2
1	F4
2	F7
3	X
4	X
5	X
6	F0

Page Table of P2

- <process_id,page-number,offset>
- Each table millions entry
- Control bit
- Increases search time,
- Chained Pointer - Shared memory

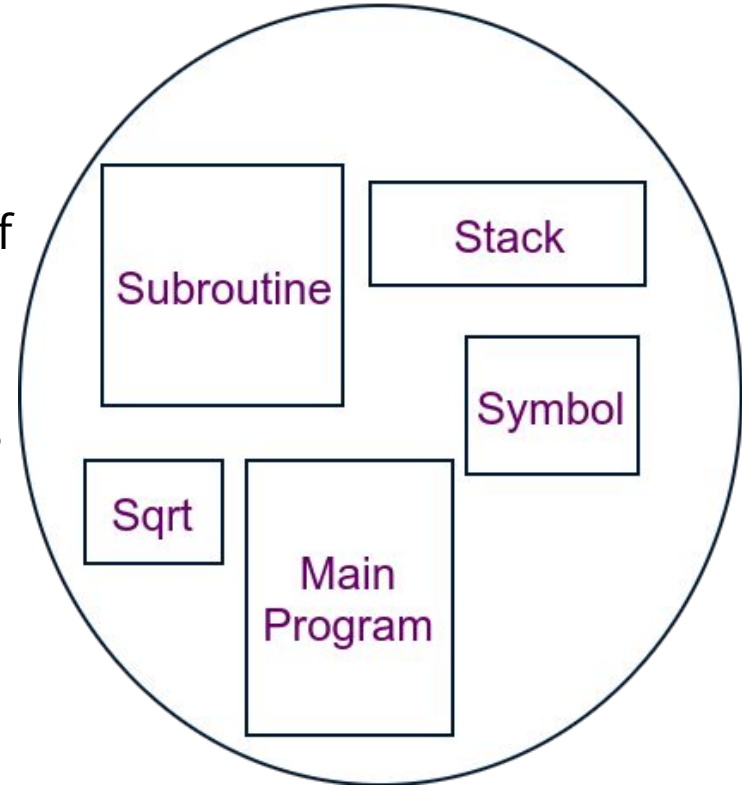
Hashed Page Table

- In a hashed page table, the virtual addresses are hashed into the hash table.
 - The virtual Page Number (which is the hash value).
 - The value of the mapped page frame.
 - A pointer to the next element in the linked list.
- Clustered Page Table



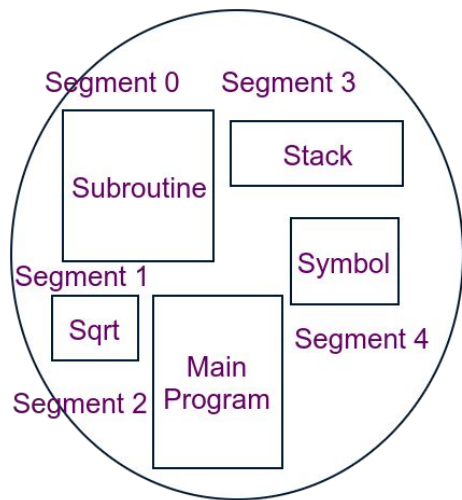
Segmentation

- Paging - separation of the user's view and physical memory
- User prefer to view memory as a collection of variable sized segments with no necessary orders
- Segmentation is scheme that supports user's view. A logical address space -> collection of segments -> a name and length
- Logical address = <segment-number, offset>
- Compiler automatically construct segments
 - Code, Global Variables, Heap, Stack, Standard Lib
- Linker -> linked to assigned segments
- Loader take all segment and assign number



Segmentation: Hardware

- User refer as 2d address where memory 1D
- Segment Table - map them
 - Segment base and segment limit
- Addressing error -> Check limit, offset



	limit	base
0	1000	1400
1	400	6300
2	400	4300
3	1100	3200
4	1000	4700

